



Forecasting the Impact of Anomalous Events on Business Process Performance

Avirup Saha
IBM Research
India
avirup.saha2@ibm.com

Neelamadhav Gantayat
IBM Research
India
neelamadhav@in.ibm.com

Sambit Ghosh
IBM Research
India
sambit.ghosh@ibm.com

Renuka Sindhgatta
IBM Research
India
renuka.sr@ibm.com

ABSTRACT

The performance of business processes such as order fulfillment, is measured and monitored in terms of Key Performance Indicators (KPIs) measured at periodic time intervals often hourly, daily, or weekly, depending on the business process. A business process is enabled by a set of IT software systems, infrastructure components, and external resources. Often, the impact of an anomalous event, such as a failure of an underlying IT system or poor weather conditions, on the business KPIs is unknown. Lack of knowledge as regards the impact of anomalous events leads to delays in handling these events. Providing a prior assessment of the impact can help prioritize the corrective actions and minimize further effect. In this work, we present a system that first uses periodic measurements of the business KPIs as a time series to correlate and identify the events impacting the KPIs, and forecasts the impact of anomalous events on the process performance. Our approach considers various types of process tasks that characterise a business process execution. We present experiments on synthetic and real-world datasets to validate the effectiveness of our approach in identifying and forecasting the impact of anomalous events on Business KPIs. We have deployed this functional component as a part of an IT Operations platform.

CCS CONCEPTS

• **Applied computing** → **Business-IT alignment**; *Enterprise computing infrastructures*; *Business process monitoring*.

KEYWORDS

key performance indicators, time series forecasting, IT anomalies

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1 INTRODUCTION

The execution of business processes such as order approvals for procurement, admission, or discharge of patients in a hospital follows an ordered sequence of tasks that need to provide a product or a service [11]. The efficiency and the effectiveness of the execution are measured by a set of quantifiable metrics known as key performance indicators (KPIs). Depending on the business process different KPIs such as the *cycle time* (the time for the complete execution of the process) or *cost of execution* are defined and measured periodically at different time intervals (such as hourly, daily, or weekly). Forecasting these business KPIs enables proactive identification of the inability to meet the goals further allowing the business process owners to take suitable actions.

A business process execution relies on the underlying software systems that include user-interfacing applications, workflow engines, and enterprise management systems. With the advent of the Cloud, many organizations have parts of their process executing on cloud-native systems while others are on the organizations' own infrastructure. Hence, current-day business processes rely on complex software ecosystems comprising multiple applications with high levels of cross-dependencies, heterogeneity, and redundancies. The applications are also built and operated on highly elastic and shared infrastructures. Figure 1 illustrates a sample business process that processes orders submitted by users. The execution of the process is supported by various services, message streaming platforms, workflow engines, and enterprise resource planning systems. Further, each of these are hosted on different infrastructure elements. This ever-growing complexity of the software ecosystem exacerbates the efforts to manage business process executions. Often failures in the software applications or infrastructure have an impact on the business process KPIs.

Business process monitoring tools enable continuous observations of business process KPIs measuring cost, time, and quality [11]. There are many application monitoring tools that provide continuous monitoring to detect anomalies, and failures and diagnose performance issues [19]. However, the monitoring of the business process and the underlying applications and infrastructure occur in silos. Hence, the impact of the failure of an application, or other anomalous events such as bad weather, on the business KPI (e.g.

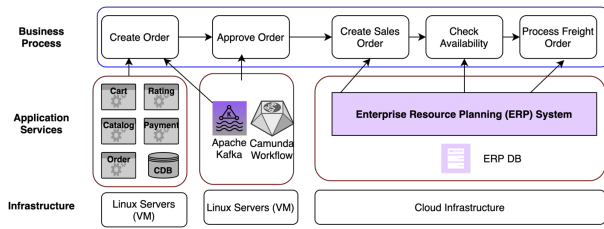


Figure 1: Business process and the underlying heterogeneous software application services and infrastructure elements

lost orders in an order processing system or delays to the goods delivery) is often unknown. Lack of information on the impact of an anomalous event on the business KPIs often leads to incorrect prioritization of anomalies, delayed diagnosis, and poor customer experience. As the KPIs and IT anomalies are monitored periodically, we pose it as a time series forecasting problem. Business KPIs such as the average number of orders processed per hour are measured periodically by a business monitoring tool. Similarly, IT metrics such as the utilization of Disk Space can be periodically measured. Hence, we collect the time series data from both business and IT monitoring tools with the objective of proactively forecasting the impact of an IT anomaly on a business KPI.

In the context of the problem, there are some requirements that need to be addressed. First, the occurrence of IT anomalies is infrequent and hence the time series data is sparse. Second, the anomalies often cause changes in the trends of the business KPIs which need to be forecasted. Here, the suitability of forecasting models based on commonly reported performance metrics such as mean absolute error is insufficient. Finally, as there can be a large number of IT anomalies captured by the application monitoring tools as compared to the business KPIs, the approach needs to provide explanations on the anomalies impacting the KPIs to ensure spurious correlations are not considered during forecasting impact. We consider the above challenges and propose a system that transforms the data obtained from different monitoring tools as a time series and identifies the influence of anomalous events on the business KPIs. Further, the forecast of business KPIs with a focus on identifying changes in trend quantifies the impact of an anomalous event. We establish the usefulness of our system by validating the following in practice:

- We propose an automated impact identification and prioritization system that considers business KPIs and other anomalous events as input. The proposed system assists process owners to prioritize the resolution of an anomalous event by projecting its impact on the business KPIs in the near future. It also provides insights as to the possible cause of an observed undesirable change in the business KPIs by identifying influencing anomalies. It is currently deployed as part of the IBM Digital Operations Suite.
- We investigate the effectiveness of diverse time series forecasting models that include conventional statistical and machine learning methods and state-of-the-art deep-learning based models.

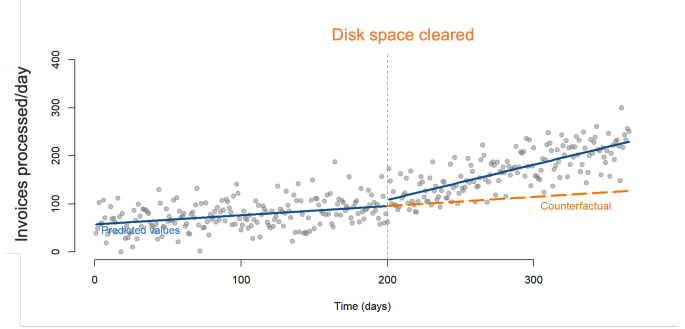


Figure 2: Low disk space monitoring in an IT monitoring tool

Existing IT monitoring and observability platforms track IT metrics and generate events (or alerts). An example of a metric is disk space is low and there will be a corresponding event when this metric is above or below a threshold. These alerts lead to the creation of incidents that are resolved by IT Operations teams. Currently, there is no understanding of how these IT operations impact the business KPIs. IT teams resolve the problems when alerts are triggered based on specific IT SLAs. However, they neither know the Business impact of IT problems nor prioritize the resolution of the problems based on Business impact KPIs. The goal is to develop a system that provides causal and counterfactual impact analysis of IT support on business KPIs.

The effect of low disk space on invoices processed per day is shown in Figure 2. We would like to answer the following questions:

- If high disk usage is left as-is, how would it impact the business KPI?
- If the IT operations took 6 hours to clear the disk space instead of 4 hours, how would it impact the business KPI?

Challenges to realizing such a system include:

- Historical data suggesting the effect of IT or environmental events on business KPIs is very difficult to get for training.
- Even though we may procure the data, there are some IT or Environmental events that are rare in nature, and we may not have our model trained on such events
- One model may not be enough in case the data is skewed.
- There are challenges pertaining to online and zero-shot learning, when the IT event has not been previously encountered by the model.

2 RELATED WORK

We position our work with respect to the existing work, categorized into the following two topics:

2.1 Business Process Performance Monitoring

Early work on process performance metrics and KPIs proposed a framework to monitor the process performance metrics and additional quality of service (QoS) metrics (availability or response time of a process engine) with the objective of identifying the main factors that influence the adherence or violation of the process performance. The authors use a decision tree to identify the influencing

factors [31]. Additionally, there has been work on categorizing process performance KPIs into different performance perspectives such as cost, time, and quality [11]. In the context of monitoring KPIs of applications (Page views, Search calls, and the number of users), recent work proposes a method for root cause analysis (RCA) [34]. The approach addresses KPIs that are aggregated or derived by complex calculations based on fundamental metrics from the raw data. RCA aims to identify the reason for an anomaly in the KPI based on fundamental metrics. Here, there are limited temporal dependencies between the fundamental metrics and the KPIs. Similar work has been done in the context of detecting anomalies in the seasonal KPIs of Web applications [28]. However, monitoring business and IT (or application) KPIs and identifying their dependencies and impact has not been considered.

2.2 Multivariate Time series

Analysis and prediction of multivariate time series (MTS) data has been one of the key areas under investigation and has been applied in a wide variety of practical problems including finance [5], covid-19 cases [29], and oceanic variables [26]. Multivariate time series models can be broadly classified into two categories.

2.2.1 Statistical models. The first category includes statistical models like VAR [24], ARIMA [24], and ARIMAX [24]. These statistical models were applied to a diverse set of problems, including wind speed [17], temperature [10], stock market estimates [16], and more.

2.2.2 Neural network models. The second category is based on neural networks [4]. The initial version of neural network-based time series predictions uses recurrent networks and Artificial Neural Networks (ANN) [3, 18, 23]. Subsequently, a long short-term Memory based encoder-decoder (LSTM-ED) framework has been proposed to capture the long-term dependencies between the parameters [20]. Next, a temporal fusion transformer that combines high-performance multi-horizon forecasting with understandable insights into temporal dynamics has been proposed [22]. To exploit the latent spatial dependencies between pairs of variables, graph neural networks (GNNs) were introduced [33]. Then to capture fine-grained geographical and temporal correlations in the data an Adaptive Graph Convolutional Recurrent Network (AGCRN) has been suggested [2]. Additionally, there has been analysis on temporal conditional Variational AutoEncoder (VAE) for MTS data with distributional drift adaptation [15]. In order to extract dynamic internal correlations, the PSTA-TCN framework has been introduced which combines stacked TCN backbones with a parallel spatiotemporal attention mechanism to extract features from various window sizes [13]. Again “LightTS”, a light deep learning architecture based on simple MLP-based structures has also been used for MTS forecasting [37]. Recently, to overcome the problems of the volatility of the predictions and their computational complexity, a simple model called “N-HITS”, that employs unique hierarchical interpolation and multi-rate data sampling strategies was proposed [6]. The datasets used in the above papers are mostly related to traffic [2, 15, 22, 37], energy [15, 37], electricity [15, 22, 37], and data collected from sensors [13, 37] which consist of repetitive patterns. There have also been efforts to apply large foundation models to time series forecasting, such as Autoformer [32], Informer [39], Fedformer [40],

and PatchTST [27]. Such models require a large amount of training data, which is not readily available in our use case.

To the best of our knowledge, we are the first to examine the issue of forecasting how anomalies would affect the functionality of business operations.

3 PROPOSED SYSTEM

The components of our proposed framework are shown in Figure 3. The input to our system is data from monitoring tools. Business process monitoring tools capture various activities (or tasks) getting executed as a part of the business process. These are stored into *event logs*. The event logs have details of each business event containing information of the activity (or task), its completion, and the specific case (such as the order, invoice, or user). The event logs are transformed into a time series of business KPIs. Similarly, the tools monitoring application and infrastructure capture common signals such as *latency*, *error*, and *traffic*. The data is transformed into a time series of application and infrastructure metrics.

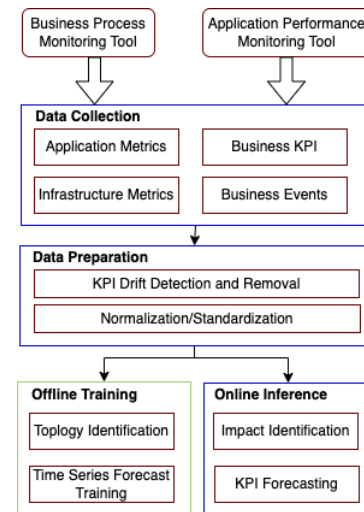


Figure 3: System overview

3.1 Data collection and pre-processing

In some cases, the business data is not readily available in a time series format. Often only the raw event log is available from Business Intelligence systems such as SAP¹ where the event name along with the end timestamp is available (along with other metadata). We construct KPIs from these business event logs by dividing the time into fixed-length intervals and counting the number of events of each type occurring (more precisely, finishing) in an interval. In this way, we obtain a single time series representing a KPI for each event type, which may be interpreted as a throughput KPI for the corresponding business process step. More complex KPIs (such as *average order cycle time* which computes the average time taken by an order to reach the end of the process flow) are usually computed by a business process mining system which can analyze the event log such as IBM Process Mining² and are available as a

¹<https://www.sap.com>

²<https://www.ibm.com/products/process-mining>

data stream. In case such a process mining system is not available, we need to perform computations using basic KPI definitions to obtain the values of the KPI at predefined time intervals.

In the case of IT events, we are dependent on IT observability tools³ to track the IT events as they occur and record them in logs. These events would capture the infrastructure, platforms, and applications. Generally, both the start and stop times of the IT events are available from the IT observability logs. We encode the IT events of a specific type into a suitable signal representation e.g., when an IT event is active, we set the corresponding signal value to +1 and when it is deactivated, we set the signal value to 0. In this way, we obtain a single time series for each IT event type.

Environmental metrics, such as weather metrics are obtained from a weather monitoring service such as the IBM Environmental Intelligence Suite (EIS)⁴ which provides APIs for retrieving historical weather data at specific geographical locations. The weather metrics are usually available as a continuous data stream from IBM EIS APIs and so no preprocessing is required for them.

3.2 KPI drift detection and removal

KPI concept drift (i.e. a change in the underlying distribution of the KPI) [14, 25] is a common problem in business processes, and some KPIs exhibit marked periodicity profiles [38]. KPI concept drifts can pose a serious challenge to anomaly detection [35]. Since we want our models to accurately forecast anomalous impacts of IT and Environmental events on the KPIs, we must account for KPI concept drift. Once data is available in the form of a multivariate time series, we use the PELT algorithm [21] to identify the change points in KPIs. We then select the largest time window or “band” with no change points in any KPI for training the forecasting models. Alternatively, the latest band with the current KPI trends can be selected for training the models. If the concept drift is expected to be temporary, the former option is preferable, whereas if it is expected to be permanent, the latter option is to be used.

3.3 Offline training

This includes training topology identification and time series forecasting components of the system.

3.3.1 Topology identification. We identify causal relationships between IT and Environmental events and the KPIs using the PC-stable algorithm [9]. We thus obtain a “Business-IT topology” where we can map IT (and Environmental) anomalies to business KPIs. This topology enables IT Site Reliability Engineers (SREs) to narrow down on specific IT anomalies for remediation based on business impact. Since the number of IT anomalies may be very large, this is quite helpful in practice. Also, since it may be difficult to train downstream time series forecasting models on data with high dimensions, this step provides valuable dimension reduction which enables these models to perform better by reducing the noise in the data. Details are given in Section 5.1.

3.3.2 Time series forecast training. We train a variety of statistical, machine learning, and deep learning multi-step forecasting models for forecasting the future values of KPIs in presence of an IT or

Environmental event at multiple horizons based on a fixed lag k (set to 24 in our context). The forecasting models are trained on the data obtained after dimension reduction based on pruning the IT anomalies which have no causal relationships with the KPIs. We evaluate various statistical, machine learning, and deep learning based models for this task and deploy the best performing one for the specific scenario. Details are given in Section 5.2.

3.4 Online Inference

The following components of the system infer the impact of an anomaly and identify the process steps which are under threat.

3.4.1 KPI impact forecasting. We use our models to forecast the KPI values in presence of the IT and Environmental events. The forecasts from the models are shown on the dashboard used for deployment (Figure 8). From there the dashboard user (usually a domain expert) can identify the impacted KPIs and quantify the KPI impact based on domain knowledge.

3.4.2 Identification of at-risk process steps. Since some KPIs are merely throughput KPIs that measure the throughput of a process step, it is possible to identify the potential impact of an IT or Environmental event on a process step by observing the forecasted values of the corresponding throughput KPI. If the dashboard user finds the health of a process step to be at risk based on a significant drop in the forecasted value of the corresponding throughput KPI, the Subject Matter Experts (SMEs) responsible for that process step (usually a separate microservice) can be alerted. In this way, our system can enable businesses to pre-emptively identify and remediate a problem in a specific business process step before it actually materializes, thereby avoiding loss of revenue.

4 DATASETS

Here we describe the datasets used in the paper.

4.1 SO data

We acquired data from a real-world customer deployment for our experimentation. Data pertaining to our client is related to the sales order process which involves creating sales orders for products or services⁵. The process supports creating orders, creation of outbound deliveries, posting goods issues, and creation of invoices as shown in Figure 4. This data set is very limited both in terms of duration and the number of data points. We have 1100 data points with 5-minute time intervals spanning roughly four days. A sample sales order process is shown in Figure 4 where the steps marked with an asterisk correspond to this data. Hereafter, this data is referred to as “SO data”.

The characteristics of the data are as follows:

- **KPIs:** ‘SALES DOCUMENTS THROUGHPUT’, ‘SD BILLING THROUGHPUT’
- **IT metrics:** ‘Batch Job Error’, ‘qRFC Error’

The KPI *SALES DOCUMENTS THROUGHPUT* records the throughput of the *Create Order* step while the KPI *SD BILLING THROUGHPUT* records the throughput of the *Post Goods Issue* step.

³<https://www.instana.com>

⁴<https://www.ibm.com/in-en/products/environmental-intelligence-suite>

⁵https://help.sap.com/docs/SAP_BUSINESS_BYDESIGN/2754875d2d2a403f95e58a41a9c7d6de/2cee28f1722d101488a1b9611201520c.html?version=2202

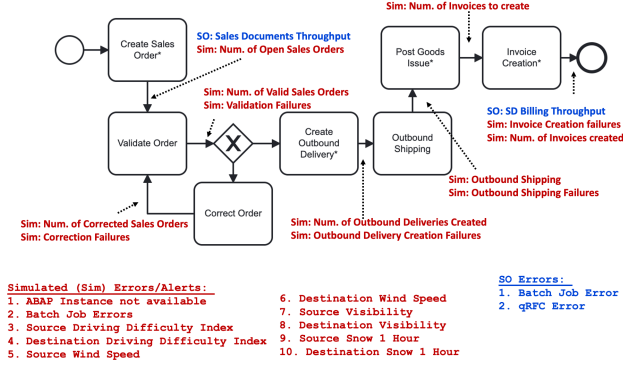


Figure 4: Overview of Sales Order (contains the steps marked with *) & Simulated processes

4.2 Simulated data

As the SO data is limited for deep learning models to train, we simulated the process in a controlled environment based on the sales order process with additional order validation, order correction, and outbound shipping steps as shown in Figure 4. We obtained 20475 data points at 1-hour time intervals spanning over 2 years. This data is representative of sales order data from our second client but is more complete. We refer to this data as “Simulated data”. Here the orders are simulated along with the IT errors. We also include an assortment of weather metrics from IBM Environmental Intelligence Suite (EIS)⁶. We design the simulation so that poor weather is likely to affect the outbound shipping step.

The characteristics of the simulated data are as follows:

- **KPIs:** ‘Correction Failures’, ‘Goods Issue Creation Failure’, ‘Invoice Creation Failures’, ‘Number of corrected sales orders’, ‘Number of invoices created’, ‘Number of invoices to create’, ‘Number of open sales order’, ‘Number of outbound deliveries created’, ‘Number of valid sales orders’, ‘Outbound Delivery Creation Failures’, ‘Outbound Shipping’, ‘Outbound Shipping Failures’, ‘Validation Failures’
- **IT metrics:** ‘ABAP Instance not available’, ‘Batch Job Errors’
- **Weather metrics:** ‘source drivingDifficultyIndex’, ‘source snow1Hour’, ‘source visibility’, ‘source windSpeed’, ‘destination drivingDifficultyIndex’, ‘destination snow1Hour’, ‘destination visibility’, ‘destination windSpeed’

The KPIs *Correction Failures*, *Validation Failures*, *Goods Issue Creation Failure*, *Invoice Creation Failures*, *Outbound Delivery Creation Failures* and *Outbound Shipping Failures* record the number of failures in the *Correct Order*, *Validate Order*, *Post Goods Issue*, *Create Invoice*, *Create Outbound Delivery*, and *Outbound Shipping* respectively. The KPIs *Number of open sales order*, *Number of valid sales orders*, *Number of corrected sales orders*, *Number of outbound deliveries created*, *Outbound Shipping*, *Number of invoices to create*, *Number of invoices created* measure the throughput of the *Create Sales Order*, *Validate Order*, *Correct Order*, *Create Outbound Delivery*, *Outbound Shipping*, *Post Goods Issue* and *Create Invoice* steps respectively.

⁶<https://www.ibm.com/in-en/products/environmental-intelligence-suite>

Table 1: Dataset statistics. “Pruned” means the dataset after pruning anomalies in the topology identification step. “GT” means the dataset with only ground truth anomalies.

Dataset	#Obs	Variables	KPIs	IT/Env
SO (Original)	1100	27	2	25
SO (Pruned)	1100	6	2	4
SO (GT)	1100	4	2	2
Simulated (Original)	20475	45	13	32
Simulated (Pruned)	20475	24	13	11
Simulated (GT)	20475	23	13	10

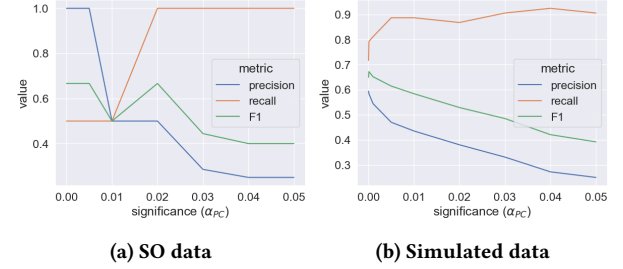


Figure 5: PC-stable performance

4.3 Topology identification

We run the topology identification step on both the SO and Simulated datasets, resulting in three separate versions of each dataset: (1) the original version with all anomalies, (2) a “pruned” version with only the anomalies identified as having a causal relation with at least one of the KPIs in the topology identification step, and (3) a “ground truth” version of the same as per domain experts. Although we experiment on all three versions, we point out that in some cases it may not be feasible to train forecasting models on the original data due to the curse of dimensionality, while in other cases the ground truth information may not be available. Table 1 provides the total number of observations, time series variables, KPIs, and IT or environmental metrics for each dataset.

5 EXPERIMENTS AND RESULTS

In this section, we discuss our experiments on topology identification and time series forecasting.

5.1 Topology Identification

In this subsection, we discuss the inferred topology of the datasets obtained using various techniques. This topology is useful in identifying which IT and Environmental metrics affect which KPIs. In this analysis, we ignore the dependence relationships between KPIs since such relationships have been extensively studied in prior art [1]. We experimented with the PC-stable algorithm [9] with varying significance levels α_{PC} . The implementation of this algorithm is available in the *tigramite*⁷ library.

The precision, recall, and F1 scores obtained using the PC-stable algorithm are shown in Figure 5. We chose the significance level

⁷<https://github.com/jakobrunge/tigramite>

Table 2: Forecasting results on all datasets

Dataset	Model	Original				Pruned Anomalies				Ground Truth Anomalies			
		MAE	RMSE	MAPE	PCC	MAE	RMSE	MAPE	PCC	MAE	RMSE	MAPE	PCC
SO	VAR	1.06E+61	1.57E+30	8.03E+60	-0.0011	24.42	6.95	42.12	0.0086	5.83	2.73	0.68	0.1909
	XGBoost	5.58	2.70	0.66	0.1932	0.28	0.67	0.78	0.0153	6.14	2.83	0.70	0.1621
	AGCRN	5.26	2.56	0.67	0.0034	0.32	0.71	0.99	0.0005	5.28	2.57	0.67	0.0063
	MTGNN	9.23	3.34	0.87	0.0010	3.04	1.74	4.24	0.0676	6.25	2.86	0.58	0.0128
	Recurrent-GRU	5.62	2.61	0.82	-0.0112	0.28	0.67	0.78	0.0153	6.14	2.83	0.70	0.1621
	DecoderMLP	5.46	2.61	0.74	-0.0094	0.15	0.60	0.37	0.5589	5.46	2.61	0.73	0.0089
	TFT	5.45	2.62	0.71	-0.0070	0.14	0.59	0.34	-0.1567	5.45	2.61	0.67	0.0216
	N-HITS	5.52	2.65	0.70	0.2316	0.11	0.52	0.25	0.0388	5.93	2.75	0.70	0.2126
Simulated	VAR	10.95	4.72	0.96	0.1603	12.34	5.27	0.87	0.2423	10.75	4.68	0.93	0.1690
	XGBoost	11.04	4.97	0.98	0.1714	13.28	5.88	0.73	0.3245	11.27	4.95	1.00	0.1798
	AGCRN	11.04	4.66	0.98	0.0590	12.71	5.23	0.96	0.0921	8.39	4.99	0.88	0.2457
	MTGNN	12.98	4.95	1.16	0.0757	22.43	6.69	1.39	0.1216	14.94	5.00	1.50	0.0925
	Recurrent-GRU	9.02	4.81	0.96	0.0970	10.86	5.37	0.98	0.0897	9.90	4.68	0.91	0.1214
	DecoderMLP	9.04	4.87	0.80	0.0006	10.23	5.45	0.88	0.0000	9.03	4.88	0.80	0.0007
	TFT	8.88	4.92	0.90	-0.0010	9.93	5.45	0.79	0.3564	8.42	4.90	0.87	0.2814
	N-HITS	41.72	13.42	4.23	0.0492	11.05	5.48	0.97	0.1109	9.46	4.90	0.89	0.1166

Table 3: Forecasting results on SO Data difficult subset (see Section 5.2.4 for details)

Model	MAE	RMSE	MAPE	PCC
VAR	4.67	5.90	0.62	0.6083
XGBoost	4.63	6.85	0.60	0.5237
AGCRN	5.86	7.12	0.72	0.2284
MTGNN	6.20	7.88	0.59	0.2227
Recurrent-GRU	2.70	2.04	0.88	-0.0809
DecoderMLP	2.56	2.02	0.77	0.2701
TFT	2.45	1.95	0.67	0.1009
N-HITS	3.61	2.43	1.02	0.2125

with the highest F1 score, with preference given to higher recall in case of a tie. Hence for SO data we chose the significance level $\alpha_{PC} = 0.02$ while for Simulated data we chose $\alpha_{PC} = 0.0001$.

5.2 Forecasting

In this subsection, we discuss the overall results obtained by using various multivariate time series (MTS) forecasting techniques on the datasets. We also describe the MTS forecasting techniques used, the experimental setup, and the metrics used for evaluating the performance of the models.

5.2.1 Forecasting techniques. We have used the following forecasting techniques:

- **VAR**, short for Vector Autoregression [24], is a statistical model for MTS forecasting. The model implementation available from the *statsmodels library*⁸ is interpretable and offers in-built support for statistical tests such as Granger Causality and Impulse Response analysis. We use this implementation in our experiments.
- **XGBoost** [7], short for eXtreme Gradient Boosting, is a machine learning algorithm based on tree boosting, which

we use on the regression task in our context. We use the python version of XGBoost Regressor⁹, further augmented with the scikit-learn¹⁰ Multi Output Regressor wrapper interface which fits one regressor per target variable. This allows us to perform MTS forecasting by regressing on the past values of the variables.

- **AGCRN** [2] is a GCN based method for MTS forecasting, developed especially for the traffic forecasting problem. It automatically infers the location graph from the time series data, which makes it interpretable. We have used the original implementation given by the authors.
- **MTGNN** [33] is a GNN based method for MTS forecasting. It automatically infers the graph connecting the time series variables, which makes it interpretable. We have used the original implementation given by the authors.
- **Recurrent-GRU**¹¹ is a simple Recurrent Network-based model for MTS forecasting using a GRU [8] layer followed by an output layer. We have used the implementation given by the PyTorch Forecasting library¹².
- **DecoderMLP**¹³ is an encoder-decoder model which uses a Multi-layer Perceptron (MLP) that predicts output only based on the decoder information. We have used the implementation given by the PyTorch Forecasting library.
- **TFT**, short for Temporal Fusion Transformer [22] is an interpretable transformer-based model for MTS forecasting. We have used the implementation given by the PyTorch Forecasting library.
- **N-HITS** [6] is a neural model for MTS forecasting based on hierarchical interpolation which provides an interpretable non-linear decomposition. We have used the implementation given by the PyTorch Forecasting library.

⁹https://xgboost.readthedocs.io/en/stable/python/python_api.html

¹⁰<https://scikit-learn.org/>

¹¹https://pytorch-forecasting.readthedocs.io/en/stable/api/pytorch_forecasting_models.rnn.RecurrentNetwork.html

¹²<https://pytorch-forecasting.readthedocs.io/>

¹³https://pytorch-forecasting.readthedocs.io/en/stable/api/pytorch_forecasting_models.mlp.DecoderMLP.html

⁸<https://www.statsmodels.org/>

Our motivation in choosing these models was mainly interpretability (characteristic of VAR, AGCRN, MTGNN, TFT, N-HiTS) and widespread acceptance and standardization (characteristic of XGBoost, Recurrent-GRU, DecoderMLP, TFT, N-HiTS).

5.2.2 Experimental Setup. Each dataset is divided into train, validation, and test sets in the ratio of 6:2:2. This split is commonly used in the literature on multivariate time series forecasting [2, 33]. All forecasting models are trained to take as input a sequence of length 24 and generate a sequence of same length as output. Hence, in the case of SO and L2C Data, we can forecast for up to 2 hours, and in the case of Simulated Data, we can forecast for up to 24 hours. AGCRN, MTGNN, Recurrent-GRU, DecoderMLP, TFT, and N-HiTS have been trained for a maximum of 100 epochs with early stopping on a server equipped with a Tesla V100 GPU. VAR and XGBoost have been trained locally on a computer equipped with a 2.4 GHz 8-Core Intel Core i9 processor and 32 GB RAM.

5.2.3 Metrics. We used the following metrics for evaluating the models. All metrics are computed on the KPI dimensions only.

- Mean Absolute Error (MAE)
- Root Mean-Squared Error (RMSE)
- Mean Absolute Percentage Error (MAPE)
- Pearson Correlation Coefficient (PCC)¹⁴

5.2.4 Discussion. With reference to Table 2, we see that the forecasting performance of all models does not deteriorate on the pruned and ground truth versions of the datasets compared to the original. In fact, the VAR model fails to fit the original version of the SO Data due to high dimensionality. It seems that for SO data, some of the IT anomalies chosen by the PC-stable algorithm have very strong correlations with the KPIs, but are not present in the ground truth provided by the domain experts. Hence for SO data, most of the models perform significantly better on the pruned dataset compared to the original and ground truth datasets. Such “false positives” uncovered by topology identification can be used for gleaning additional insights into the system in case the ground truth data is inadequate.

We see that in the three datasets we consider, there is no clear winner among the various statistical, deep learning and machine learning models. Some models perform well in terms of some metrics on some subset of the data, but not in terms of other metrics on other subsets of the data. The question then arises which model should we trust for our deployed system, which will be able to respond well to anomalies should they occur.

For the purpose of answering the above question, we select a difficult subset from the SO Data test set consisting of two samples, one where the IT error *qRFC Error* has a prominent impact on the KPI *SALES DOCUMENTS THROUGHPUT* (Figure 6) and one where the IT error *Batch Job Error* has a prominent impact on the KPI *SD BILLING THROUGHPUT* (Figure 7). Here too, we evaluate all metrics only on these two KPIs. Results are reported in Table 3.

With reference to Table 3, we see that VAR and XGBoost obtain higher PCC than the other models even though the other error metrics are higher. This indicates that they can more accurately capture the peculiar KPI trend induced by the IT anomalies.

From Figure 6 we see that VAR and XGBoost successfully detect the trough in the KPI *SALES DOCUMENTS THROUGHPUT* in the first sample, while no other model is able to do the same. In the second sample, from Figure 7 we see that VAR, XGBoost, and Recurrent-GRU successfully detect the trough in the KPI *SD BILLING THROUGHPUT*. In this sample, XGBoost also incorrectly predicts a trough in the KPI *SALES DOCUMENTS THROUGHPUT*. From Table 3 we see that despite predicting a false positive, XGBoost obtains better MAE, MAPE and RMSE scores than VAR. However, VAR has a higher PCC score than XGBoost. Also, TFT obtains the best MAE and RMSE, while MTGNN obtains the best MAPE. These models, however, do not predict the troughs in the KPIs but simply predict a better average value of the KPI.

Hence we conclude that in such cases where the KPIs show dramatic anomalous behavior in response to IT or Environmental events, statistical and machine learning models such as VAR and XGBoost can in fact predict such behaviour better and hence obtain better PCC scores than the state-of-the-art deep learning models. Also the PCC score is more reliable than MAE, MAPE and RMSE in such cases since these error metrics are more influenced by the average value of the KPI. Since such anomalous events are of special interest to us in this work, we prefer the surety of models such as VAR and XGBoost for deployment purposes. Also, the VAR model is very interpretable and can lend itself to many types of structural analyses such as Granger Causality and Impulse Response Analysis [1, 24, 30].

6 DEPLOYMENT

The proposed system is deployed into multiple accounts, as part of the IT operations platform, to assist the site reliability engineers (SREs) supporting the applications and infrastructures. The objective is to provide insights that allow SREs to prioritize errors in the application. We created REST Application Programming Interfaces (APIs) for training and inference of both topology identification and time series forecasts. The REST APIs are deployed into a Kubernetes based private cloud platform as two different micro-services: one for topology identification and impact analysis, and the other one for forecasting. We also developed and deployed a dashboard with minimal components to capture the topology, impacted system, current KPIs, IT and infrastructural metrics, and the impact of an anomaly in case there is any.

The forecast training and inference hosted as APIs. Similarly the topology identification and impact analysis support are hosted as APIs. The forecasting models are served to scale on Kubernetes using an in-house version of the popular open-source model serving framework, *Kserve*¹⁵ to support high-volume inference API calls. In the current approach, the inference APIs are executed when the application monitoring tools capture errors. The dashboard (Figure 8) provides visibility into KPIs and metrics prior to the error along with the forecasted KPIs. The topology identification highlights the specific step in the process that is likely to get impacted based on the mapping between the throughput KPIs and the process steps. The highlighted step and the KPI forecast enable the SREs to prioritize resolution of the errors.

¹⁴<https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.pearsonr.html>

¹⁵<https://github.com/kserve/kserve>

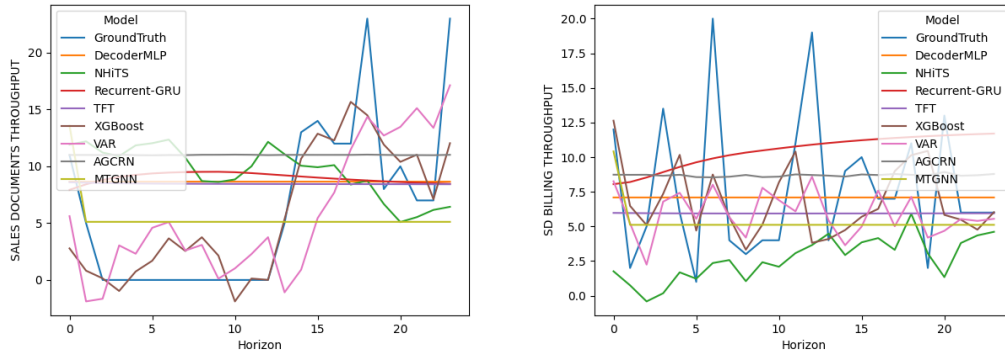


Figure 6: Forecast on SO Data difficult sample 1 (qRFC Error)

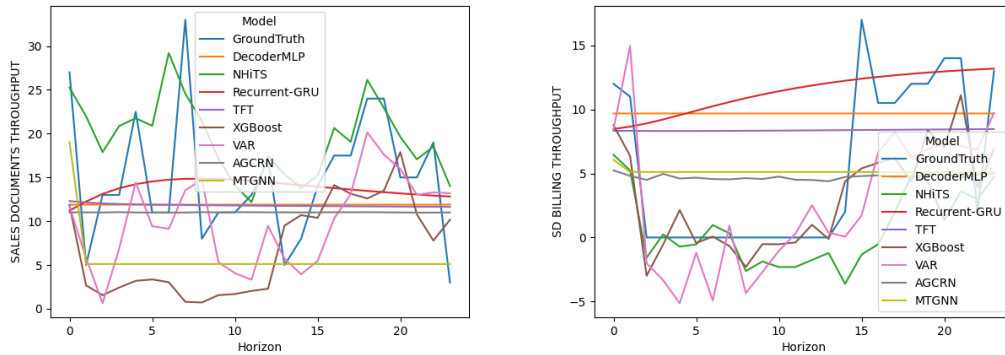


Figure 7: Forecast on SO Data difficult sample 2 (Batch Job Error)

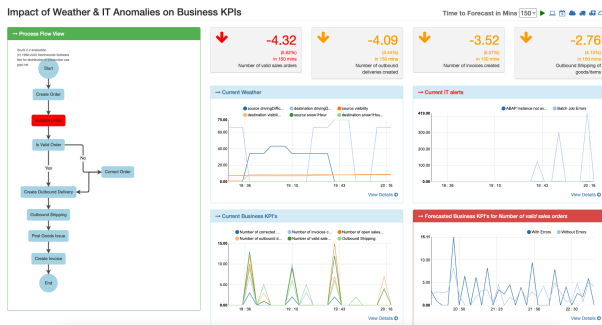


Figure 8: Dashboard showing the process view, current IT and weather metrics, and current as well as forecasted KPIs.

7 CONCLUSION AND FUTURE WORK

In this paper, we have described a first-of-a-kind system that combines Business Observability with IT and Environmental Observability. We have explored a wide range of statistical, machine learning, and deep learning models for multivariate time series forecasting to address the problem of identifying the effect of IT and

Environmental anomalies on KPIs. We have shown that the forecasting performance generally does not deteriorate on application of a causal inference algorithm for topology identification and dimension reduction. This makes our approach readily scalable to a scenario where there are dozens of KPIs and hundreds of IT anomalies, as is often the case with real-world enterprise systems such as SAP. We have also demonstrated that, contrary to expectations, statistical and machine learning models such as VAR and XGBoost can prove more useful for the current problem in limited settings than state-of-the-art deep learning models such as AGCRN and N-HITS. We have shown that, in these limited settings, PCC is a better judge of performance than standard error metrics such as MAE, MAPE and RMSE which are widely used in the literature on time series forecasting, since only a trend-based evaluation can identify whether a model accurately captures the effects of IT and environmental anomalies on KPIs. This somewhat surprising result may be due to the peculiarities of the domain. This shows that despite great advances in deep learning in recent years, these simpler models may still be relevant for niche applications. In future, we plan to explore the use of foundation models for time series [27, 36] as well as other recent state-of-the-art lightweight models such as TSMixer [12] to solve the current problem. Our data is available at <https://github.com/BizITObs/BizITObservabilityData>.

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